

THE FIFTH STATE

First working submission wins a Takealot voucher.

On Day 3 the recap slide asked: *what would you have to change to add a fifth state*, `QuoteExpired` ? This is that question, as a race.

The rules

1. Start from `lab-4-3-solution` — the app as it stands at the end of Day 4.
2. Copy `challenge/challenge_test.dart` into your app as `test/challenge_test.dart`. **Do not edit that file.** Change the app until it passes.
3. You win when **both** of these are clean, in front of the instructor:

```
flutter analyze    # No issues found!
flutter test       # All tests passed!
```

4. `flutter analyze` counts. Green tests with a warning is not a win.
 5. Every existing test must still pass. Breaking one to fix another is not a win.
 6. Ask anything you like. Use the handbooks, the solutions folder, the internet. Pair up if you want — one voucher, so agree how you'd split it first.
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What to build

A quote goes stale. Fifteen minutes after it was issued, it is no longer valid and the customer has to ask again.

1. The fifth state

In `quote_model.dart`, alongside the other four:

```
class QuoteExpired extends QuoteState {
  const QuoteExpired(this.quote);
  final Quote quote;
}
```

The moment you add it, **the app stops compiling**. That is the point. The sealed class means every `switch` over `QuoteState` now has a hole in it, and the compiler will show you exactly where. Fill them in.

2. A clock you can control

In `quote_providers.dart`:

```
final clockProvider = Provider<DateTime Function>(() => DateTime.now());
```

Nothing in the app should call `DateTime.now()` directly any more — read the clock from the provider. That is what lets the tests move time forward without waiting fifteen real minutes, and it is the same dependency-injection seam as `quoteServiceProvider`.

3. Expiry on the notifier

`QuoteNotifier` needs to remember **when** the current quote loaded, and gain:

```
void checkExpiry({Duration maxAge = const Duration(minutes: 15)});
```

Behaviour the tests check:

Situation	Result
Loaded, older than <code>maxAge</code>	becomes <code>QuoteExpired</code> , carrying the same quote
Loaded, younger than <code>maxAge</code>	stays <code>QuoteLoaded</code>
Idle, loading or failed	nothing happens
<code>reset()</code> from expired	back to <code>QuoteIdle</code>
A new <code>submit()</code> after expiry	loads, and the clock starts again

`maxAge` must be overridable — the tests pass one minute in.

Getting started

```
cd quote_app
git checkout lab-4-3-solution
git checkout -b challenge-<your-name>
cp ../challenge/challenge_test.dart test/
flutter test
```

What you will see on that first run

Compile errors, not test failures. `QuoteExpired` and `clockProvider` do

not exist yet, so the file cannot build:

```
test/challenge_test.dart:35:7: Error: Undefined name 'clockProvider'.
test/challenge_test.dart:47:19: Error: Method not found: 'QuoteExpired'.
test/challenge_test.dart:58:11: Error: 'QuoteExpired' isn't a type.
```

That list is your to-do list. Work down it.

You will also see `mini_project_flow_test.dart` **fail to load**. That is not

your fault and you have not broken anything. `flutter test` compiles the test files together, so one file that will not build takes another with it. It comes back on its own the moment `challenge_test.dart` compiles.

Sanity check at any point:

```
flutter test test/quote_notifier_test.dart    # should stay green throughout
```

Hints, if you get stuck

- The first errors are the compiler doing its job. Read them in order.
- `_loadedAt` is ordinary private state on the notifier. It does not need to be part of `QuoteState` .
- A failed quote has no load time. Clear it, or `checkExpiry` will do something surprising.
- Test 9 is the one people miss: after expiring, a **new** quote must not immediately expire as well.
- If a test fails on something you have not touched, read what it actually asserts before changing your code.

Roughly what it takes

Around **30 lines across three files**. If you are rewriting half the app, stop and re-read the spec — you have gone off the path.